

From S3 to SQL: Serverless Data Analysis Using Amazon Athena

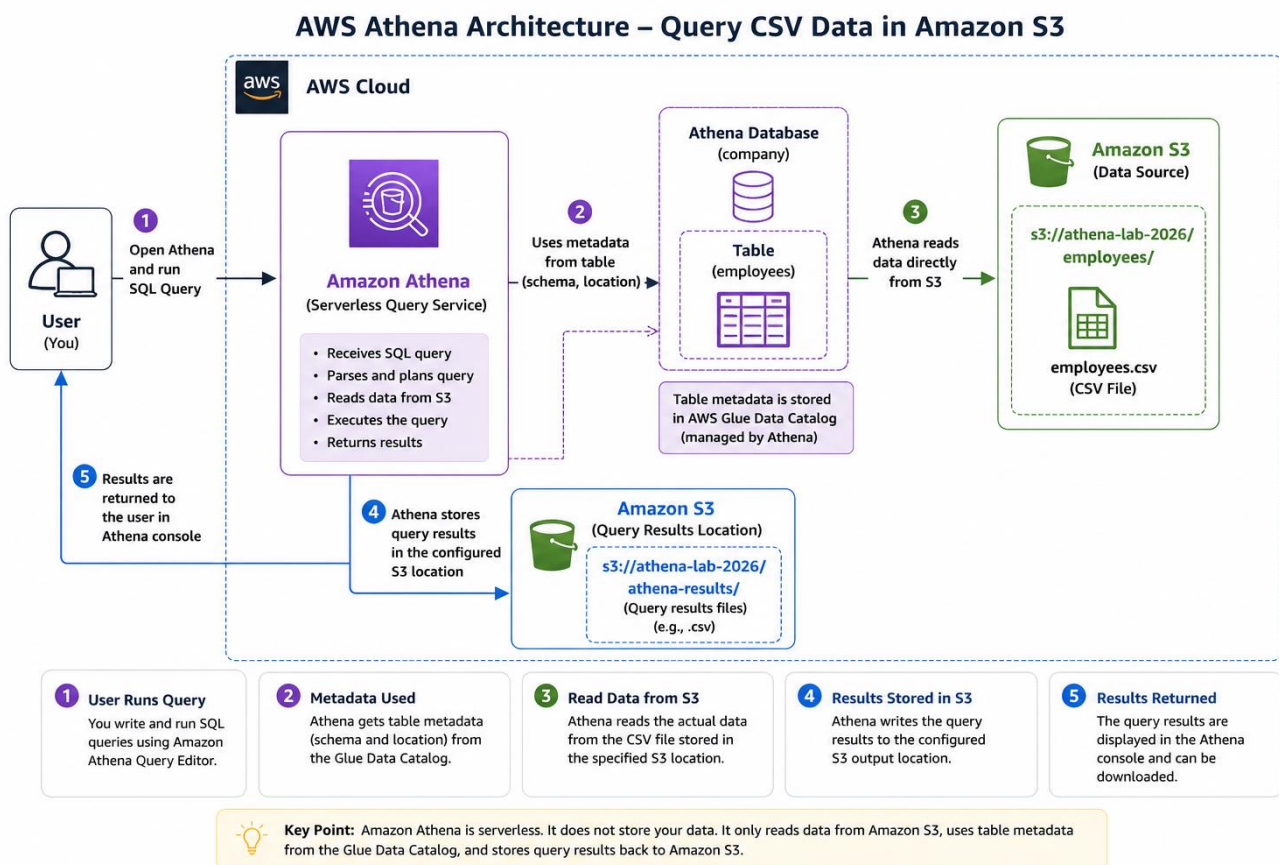
Objective:

The objective of this task is to learn how to use Amazon Athena to query and analyze data stored in an Amazon S3 bucket using SQL. This lab helps understand how Athena works without requiring a database server and demonstrates how data can be analyzed quickly using a serverless service.

Introduction:

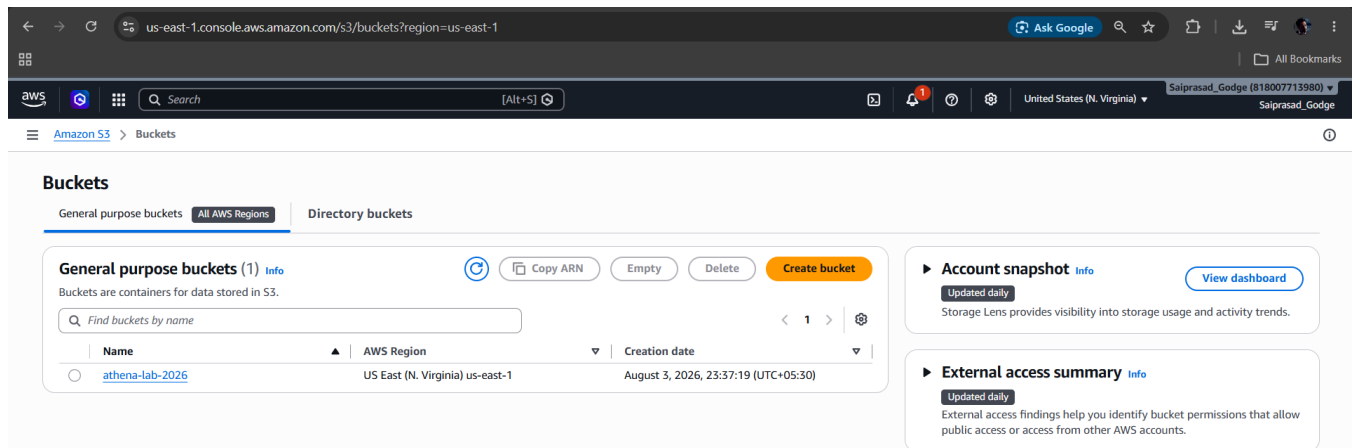
Amazon Athena is a serverless query service provided by AWS that allows users to analyze data stored in Amazon S3 using standard SQL queries. It does not require setting up or managing any database servers, making it simple and easy to use. Instead of loading data into a database, Athena reads files directly from an S3 bucket and returns the query results within seconds. It is commonly used to analyze CSV, JSON, and log files for reporting, troubleshooting, and data analysis.

Architecture Overview:



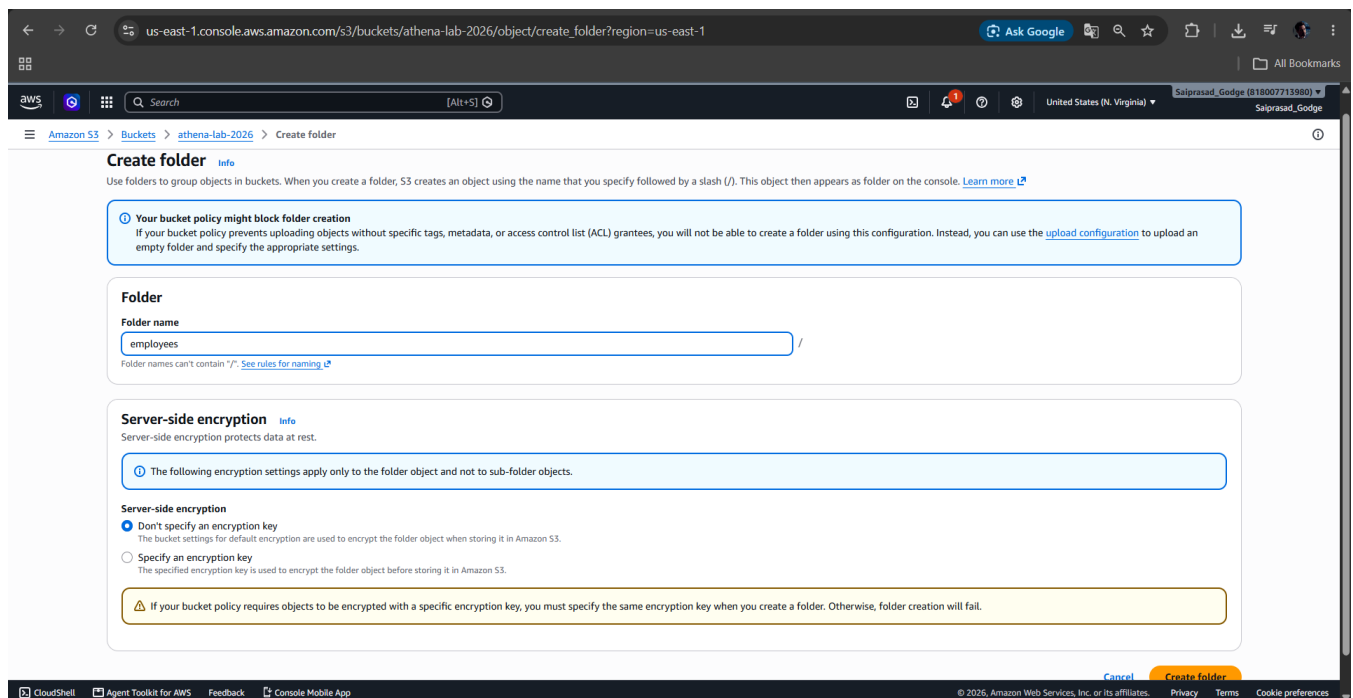
Step 1: Create an S3 Bucket

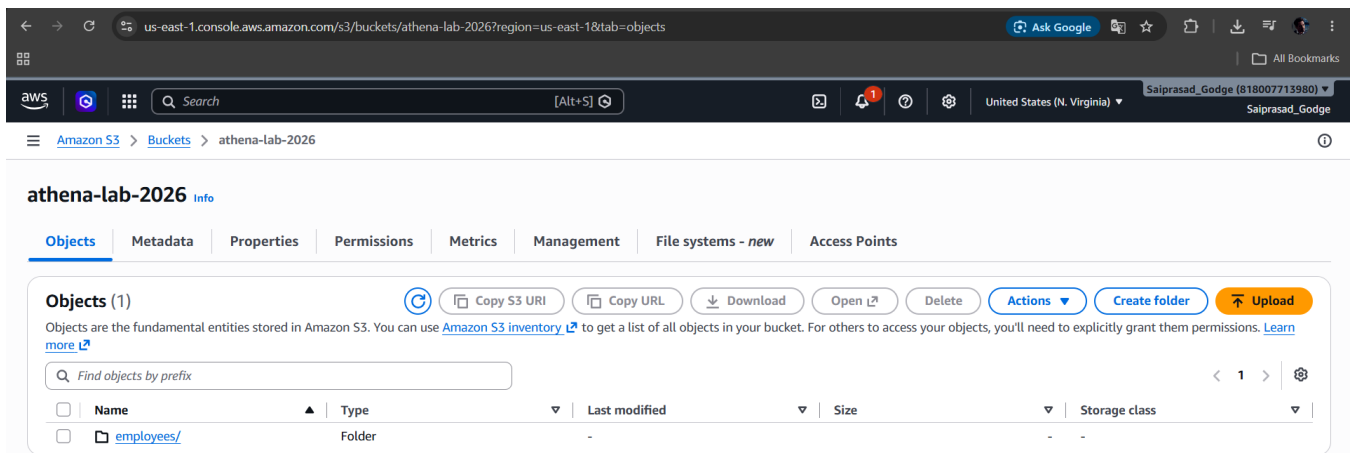
- ✓ Navigate to the **Amazon S3** console and Click **Create bucket**.
- ✓ Select **General purpose** as the bucket type.
- ✓ Choose the AWS Region (**US East, N. Virginia**)
- ✓ Enter a unique bucket name: **athena-lab-2026**.
- ✓ **Object Ownership**: ACLs disabled
- ✓ **Block Public Access**: Enabled (all blocked)
- ✓ **Bucket Versioning**: Disabled
- ✓ **Default Encryption**: SSE-S3
- ✓ Click **Create bucket**.



Step 2: Create a Folder for the Source Data

- ✓ Open the **athena-lab-2026** bucket.
- ✓ Click **Create folder**.
- ✓ Enter the folder name: **employees**.
- ✓ Keep the default encryption settings.
- ✓ Click **Create folder**.

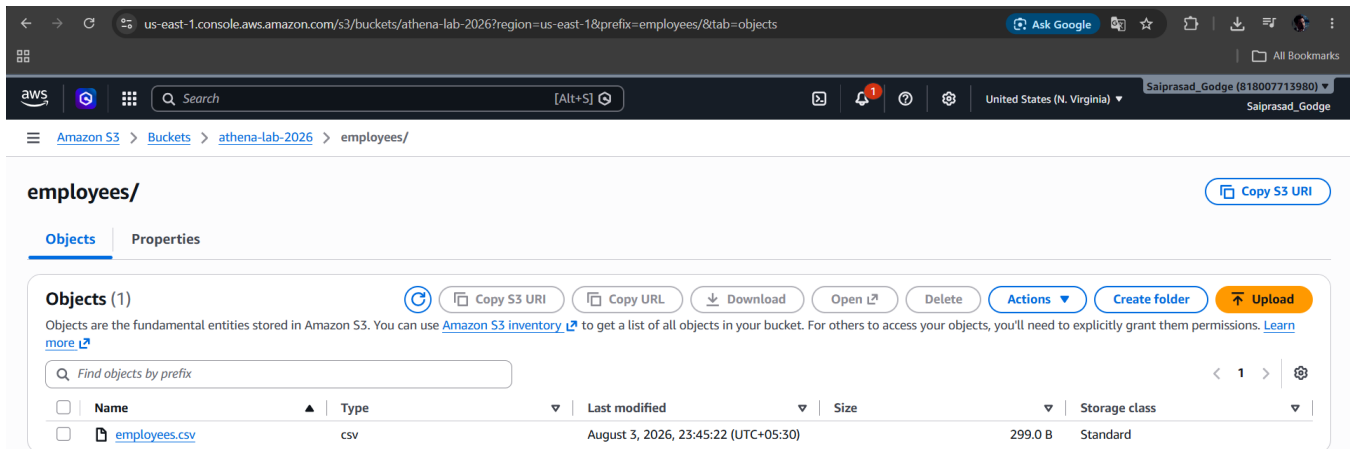




Step 3: Upload the CSV File

- ✓ Open the **employees** folder.
- ✓ Click **Upload**.
- ✓ Select the **employees.csv** file.
- ✓ Click **Upload** to add the file to the bucket.
- ✓ Verify that **employees.csv** appears in the folder.

Direct File Link: <https://github.com/SAI-GODGE/AWS-Athena/blob/main/employees.csv>



Step 4: Set the Query Result Location

Before running queries, configure where Athena should store query results.

- ✓ Open the **Amazon Athena** console.
- ✓ Go to the **Query settings** tab.
- ✓ Click **Manage** under **Query result location**.
- ✓ Set the S3 location to **s3://athena-lab-2026/athena-results/**.
- ✓ Leave the encryption and bucket owner options unchanged.
- ✓ Click **Save**.

Manage query result location and encryption

Location of query result - *optional*

Enter an S3 prefix in the current region where the query result will be saved as an object.

[View](#)

[Browse S3](#)

You can create and manage lifecycle rules for this bucket
Use Amazon S3 lifecycle rules to store your query results and metadata cost effectively or to delete them after a period of time. [Learn more](#)

[Lifecycle configuration](#)

Expected bucket owner - *optional*

Specify the AWS account ID that you expect to be the owner of your query results output location bucket.

- ☐ **Assign bucket owner full control over query results**
Enabling this option grants the owner of the S3 query results bucket full control over the query results. This means that if your query result location is owned by another account, you grant full control over your query results to the other account.
- ☐ **Encrypt query results**

[Cancel](#)

[Save](#)

The screenshot shows the Amazon SageMaker Unified Studio console. The top navigation bar includes the AWS logo, a search bar, and the user's name 'Salprasad_Godde'. The main header reads 'Query in Amazon SageMaker Unified Studio' with a 'Set up' button. Below this, there are tabs for 'Editor', 'Recent queries', 'Saved queries', and 'Query settings' (which is selected). The 'Query settings' section has a sub-header 'Query settings' and a description: 'Query settings temporarily override your workgroup configurations. Clear all the query settings to enforce your workgroup configurations.' There are two main sections: 'Query result encryption' and 'Execution controls'. The 'Query result encryption' section has four settings: 'Query result location' (s3://athena-lab-2026/athena-results/), 'Encrypt query results' (turned off), 'Expected bucket owner' (empty), and 'Assign bucket owner full control over query results' (turned off). The 'Execution controls' section has a table with columns 'Control Name', 'Value', and 'Unit'. The table lists three controls: 'Data scanned limit', 'Min DPU per query', and 'Max DPU per query', all with a value of '-' and a unit of '-'. There are 'Manage' and 'Edit controls' buttons at the end of each section.

us-east-1.console.aws.amazon.com/athena/home?region=us-east-1#/query-editor/settings

Amazon Athena > Query editor

Query in Amazon SageMaker Unified Studio [Set up](#)

Use your IAM role to analyze and build with your existing AWS resources in a single data and AI development environment.

Editor Recent queries Saved queries **Query settings** Workgroup primary

Query settings
Query settings temporarily override your workgroup configurations. Clear all the query settings to enforce your workgroup configurations.

Query result encryption [Manage](#)

Query result location	Encrypt query results	Expected bucket owner	Assign bucket owner full control over query results
s3://athena-lab-2026/athena-results/	-	-	Turned off

Execution controls [Edit controls](#)

Set limits and control query execution behavior for the workgroup. Controls apply to all queries in the workgroup unless overridden.

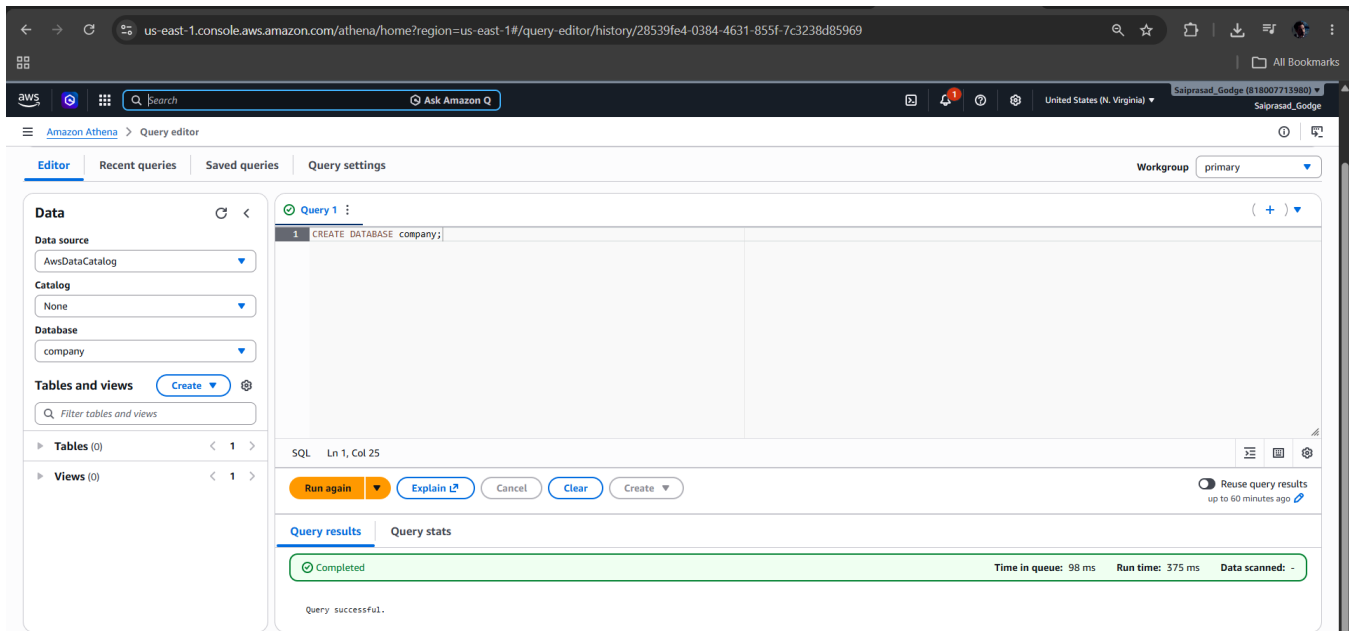
Control Name	Value	Unit
Data scanned limit	-	-
Min DPU per query	-	-
Max DPU per query	-	-

Step 5: Create a Database

In the Athena Query editor, run a CREATE DATABASE statement to create a logical container for the table:

```
CREATE DATABASE company;
```

After running the query, select company from the Database dropdown so subsequent statements target it.



Step 6: Create an External Table

Define an external table that maps to the CSV file's schema and points to its S3 location.

- ✓ Open the **Athena Query Editor**.
- ✓ Ensure the **company** database is selected.
- ✓ Enter the following SQL statement:

```
CREATE EXTERNAL TABLE employees (  
    id INT,  
    name STRING,  
    department STRING,  
    salary INT,  
    city STRING  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE  
LOCATION 's3://athena-lab-2026/employees/'  
TBLPROPERTIES (  
    'skip.header.line.count'='1'  
);
```

- ✓ Click **Run** to create the external table.
- ✓ Verify that the **employees** table appears under the **company** database.

The screenshot shows the Amazon Athena Query Editor interface. On the left, the 'Data' sidebar is visible with settings for 'Data source' (AwsDataCatalog), 'Catalog' (None), 'Database' (company), and 'Tables and views' (employees). The main editor area displays a SQL query for creating an external table named 'employees' with columns: id (INT), name (STRING), department (STRING), salary (INT), and city (STRING). The query is stored as a text file in the S3 location 's3://athena-lab-2026/employees/'. Below the query editor, the 'Query results' section shows a green bar indicating the query is 'Completed'. The query statistics are: Time in queue: 43 ms, Run time: 368 ms, and Data scanned: -.

Note:

The following steps (Step 7 to Step 14) include SQL queries executed in Amazon Athena to retrieve and analyze the employee data. The output of each query is shown below its respective step for verification.

Step 7: Query All Records

Run the following SQL query in the Athena Query Editor to confirm the table was created correctly and Athena can read every row from the underlying CSV file.

SELECT * FROM employees;

The screenshot shows the Amazon Athena Query Editor interface. On the left, the 'Data' sidebar is visible with settings for 'Data source' (AwsDataCatalog), 'Catalog' (None), 'Database' (company), and 'Tables and views' (employees). The main editor area displays a SQL query: 'SELECT * FROM employees;'. Below the query editor, the 'Query results' section shows a green bar indicating the query is 'Completed'. The query statistics are: Time in queue: 115 ms, Run time: 504 ms, and Data scanned: 0.29 KB.

Results (10)

Search rows

Copy

Download results CSV

< 1 >

#	id	name	department	salary	city
1	1	Alice	HR	45000	New York
2	2	Bob	IT	65000	Chicago
3	3	Charlie	Finance	55000	Boston
4	4	David	IT	72000	Seattle
5	5	Eva	Marketing	50000	Miami
6	6	Frank	HR	47000	Dallas
7	7	Grace	Finance	80000	Boston
8	8	Harry	IT	62000	Chicago
9	9	Irene	Marketing	53000	New York
10	10	Jack	HR	48000	Dallas

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Step 8: Filter Employees by Department

Run the following SQL query to display only the employees who belong to the **IT** department.

```
SELECT * FROM employees WHERE department = 'IT';
```

aws

Search

Amazon Athena

Query editor

United States (N. Virginia)

Saiprasad, Godge

Editor

Recent queries

Saved queries

Query settings

Workgroupprimary

Data

Data source

AwsDataCatalog

Catalog

None

Database

company

Tables and views

Create

Tables (1)

employees

Views (0)

Query 1

```
1 SELECT *
2 FROM employees
3 WHERE department = 'IT';
```

SQLLn 3, Col 25

Run again

Explain

Cancel

Clear

Create

Query results

Query stats

Completed

Time in queue: 98 msRun time: 409 msData scanned: 0.29 KB

Results (5)

CopyDownload results CSV

Search rows

#	id	name	department	salary	city
1	2	Bob	IT	65000	Chicago
2	4	David	IT	72000	Seattle
3	8	Harry	IT	62000	Chicago

Step 9: Find the Highest-Paid Employee

Run the following SQL query to display the employee who has the highest salary.

```
SELECT * FROM employees ORDER BY salary DESC LIMIT 1;
```

The screenshot shows the Amazon Athena Query Editor interface. On the left, the 'Data' sidebar is open, showing the 'Data source' as 'AwsDataCatalog', 'Catalog' as 'None', and 'Database' as 'company'. Under 'Tables and views', the 'employees' table is listed. The main editor area contains a SQL query: `1 SELECT *
2 FROM employees
3 ORDER BY salary DESC
4 LIMIT 1;`. Below the query, the 'Query results' tab is active, showing a 'Completed' status with a green bar. The 'Results (1)' section displays a table with columns: id, name, department, salary, and city. The first row shows id=7, name=Grace, department=Finance, salary=80000, and city=Boston. The 'Query stats' section shows 'Time in queue: 105 ms', 'Run time: 414 ms', and 'Data scanned: 0.29 KB'.

Step 10: Calculate the Average Salary

Run the following SQL query to calculate the average salary of all employees in the **employees** table.

```
SELECT AVG(salary) FROM employees;
```

The screenshot shows the Amazon Athena Query Editor interface. The 'Data' sidebar on the left is the same as in the previous screenshot. The main editor area contains a SQL query: `1 SELECT AVG(salary)
2 FROM employees;`. The 'Query results' tab is active, showing a 'Completed' status with a green bar. The 'Results (1)' section displays a table with a single column: `_col0`. The first row shows the value 57700.0. The 'Query stats' section shows 'Time in queue: 99 ms', 'Run time: 474 ms', and 'Data scanned: 0.29 KB'.

Step 11: Count Total Employees

Run the following SQL query to count the total number of employee records available in the **employees** table.

```
SELECT COUNT(*) FROM employees;
```

The screenshot shows the Amazon Athena Query Editor interface. On the left, the 'Data' panel shows the data source as 'AwsDataCatalog', catalog as 'None', and database as 'company'. The 'Tables and views' panel shows a table named 'employees'. The main editor area contains the SQL query: `1 SELECT COUNT(*)
2 FROM employees;`. Below the query, the 'Query results' tab is active, showing a green bar indicating the query is 'Completed'. The results table has one row with the value 10.

#	total
1	10

Step 12: Group Employees by Department

Run the following SQL query to display the total number of employees in each department.

SELECT department, COUNT(*) AS total FROM employees GROUP BY department;

The screenshot shows the Amazon Athena Query Editor interface. The SQL query is: `1 SELECT department,
2 COUNT(*) AS total
3 FROM employees
4 GROUP BY department;`. The 'Query results' tab is active, showing a green bar indicating the query is 'Completed'. The results table has four rows representing different departments.

#	department	total
1	Marketing	2
2	Finance	2
3	IT	3
4	HR	3

Step 13: Filter High Earners

Select only the name and salary of employees earning above a salary threshold:

SELECT name, salary FROM employees WHERE salary > 60000;

Query 1 :

```
1 SELECT name, salary
2 FROM employees
3 WHERE salary > 60000;
```

SQL Ln 3, Col 22

Run again Explain Cancel Clear Create

Query results Query stats

Completed Time in queue: 114 ms Run time: 440 ms Data scanned: 0.29 KB

Results (4)

Search rows

#	name	salary
1	Bob	65000
2	David	72000
3	Grace	80000
4	Harry	62000

Step 14: Verify the Bucket Structure

Returning to the S3 console and opening the athena-lab-2026 bucket now shows two top-level folders:

employees/ – Stores the source **employees.csv** file used by Amazon Athena.

athena-results/ – Automatically created by Athena to store the results and metadata of executed SQL queries.

us-east-1.console.aws.amazon.com/s3/buckets/athena-lab-2026?region=us-east-1&tab=objects

Amazon S3 Buckets athena-lab-2026

athena-lab-2026 Info

Objects Metadata Properties Permissions Metrics Management File systems - new Access Points

Objects (2)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

	Name	Type	Last modified	Size	Storage class
☐	athena-results/	Folder	-	-	-
☐	employees/	Folder	-	-	-

Step 15: Inspect the Saved Query Result Files

Open the **athena-results** folder in the S3 bucket and navigate to the latest query result directory (for example, **Unsaved** → **2026** → **08** → **04**).

Verify that the folder contains a **.csv** file and its corresponding **.csv.metadata** file for each executed query. These files are created automatically by Amazon Athena and store the query results along with the associated execution metadata, confirming that the queries were executed successfully.

04/

Copy S3 URI

Objects

Properties

Objects (17)

Find objects by prefix

Copy S3 URI

Copy URL

Download

Open L

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

	Name	Type	Last modified	Size	Storage class
☐	61dd25b2-200f-4519-8811-29db34eaa638.csv	csv	August 4, 2026, 00:11:35 (UTC+05:30)	18.0 B	Standard
☐	61dd25b2-200f-4519-8811-29db34eaa638.csv.metadata	metadata	August 4, 2026, 00:11:36 (UTC+05:30)	67.0 B	Standard
☐	679ef73b-780a-4440-924d-b3c089317bcd.txt	txt	August 4, 2026, 00:02:24 (UTC+05:30)	0 B	Standard
☐	6cb39c20-f99e-4f63-babb-b7f6ac9f6d2f.csv	csv	August 4, 2026, 00:06:48 (UTC+05:30)	409.0 B	Standard
☐	6cb39c20-f99e-4f63-babb-b7f6ac9f6d2f.csv.metadata	metadata	August 4, 2026, 00:06:48 (UTC+05:30)	238.0 B	Standard
☐	8bf2cb9-06e7-4979-be26-b9683bb30f08.csv	csv	August 4, 2026, 00:13:03 (UTC+05:30)	69.0 B	Standard
☐	8bf2cb9-06e7-4979-be26-b9683bb30f08.csv.metadata	metadata	August 4, 2026, 00:13:03 (UTC+05:30)	120.0 B	Standard
☐	c15c1b01-dbc4-4323-81f3-26dd4a7e5f09.csv	csv	August 4, 2026, 00:13:49 (UTC+05:30)	173.0 B	Standard
☐	c15c1b01-dbc4-4323-81f3-26dd4a7e5f09.csv.metadata	metadata	August 4, 2026, 00:13:49 (UTC+05:30)	111.0 B	Standard
☐	c8530ee7-d7a1-42fc-ae5a-269ff4e250f5.csv	csv	August 4, 2026, 00:10:02 (UTC+05:30)	144.0 B	Standard

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Summary:

This lab demonstrated how to use **Amazon Athena** to analyze data stored in an **Amazon S3** bucket using SQL without setting up or managing any database servers. The implementation included creating an S3 bucket, uploading a CSV file, configuring Athena, creating a database and an external table, and executing various SQL queries to retrieve and analyze the data. The query results were automatically stored in Amazon S3, confirming the successful implementation of a serverless data analysis solution using Amazon Athena.